

Understanding Accessibility Sufficiency, Activity Participation, and Quality of Life: A Study in Austin, Texas

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Introduction

- ❑ Sufficient accessibility means that all individuals have a minimum level of access to essential opportunities such as work, school, healthcare, and leisure, making it possible to meet their basic needs. However, setting an appropriate threshold for sufficient accessibility is challenging.
- ❑ To inform this threshold, researchers have examined how transport accessibility impacts people's lives by linking it to activity participation (Allen & Farber, 2020), employment opportunities (Martens et al., 2022), and social exclusion (Martens, 2016).

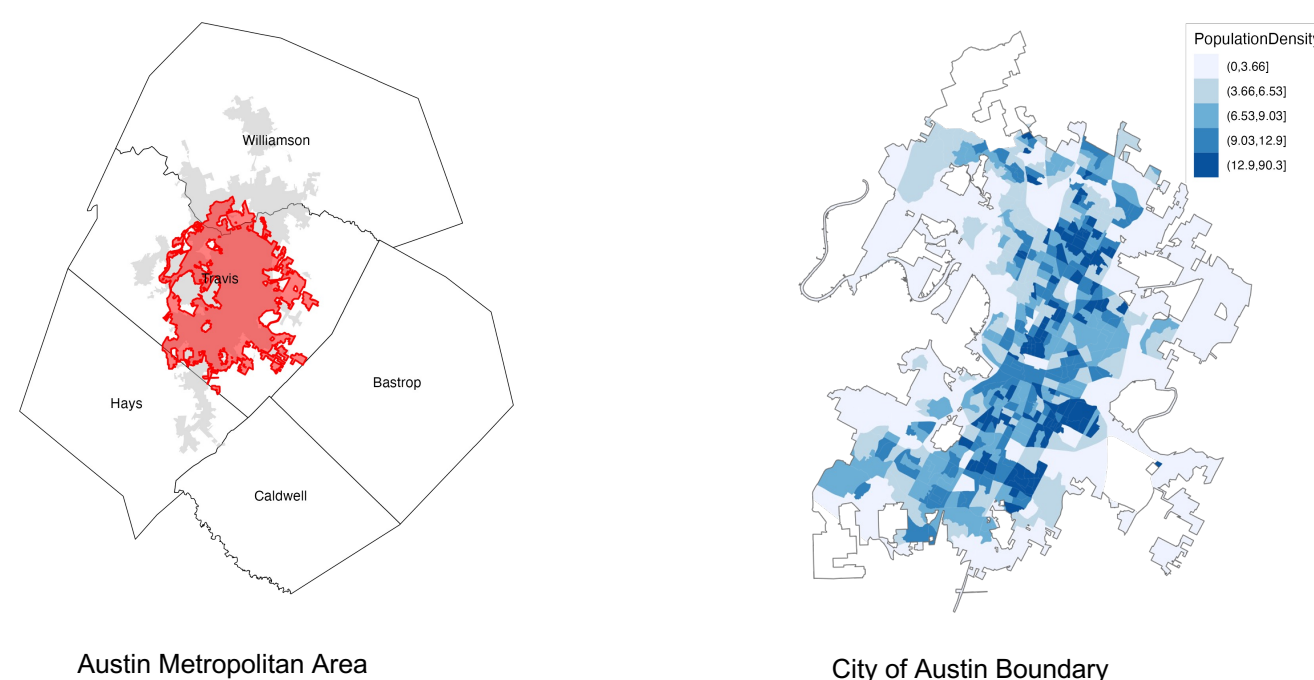
This study aims to investigate how transportation accessibility influences residents' daily activities and overall quality of life in the city, with the goal of providing empirical evidence for establishing a sufficiency threshold.

Key research questions:

- ✓ How do variations in accessibility across different areas of the city impact activity participation?
- ✓ Are there disparities in accessibility and activity participation across different demographic groups (e.g., income, racial/ethnic groups, zero car households)?
- ✓ Does the level of accessibility and activity participation impact city dwellers' self-reported quality of life (QoL)?

Methods

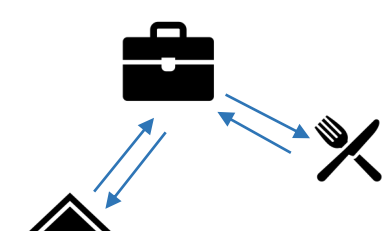
Study Area



- ❑ We measured accessibility to work (i.e., total jobs) and non-work destinations or points of interest (POI) including grocery stores and supermarkets, hospitals, urgent care, colleges and universities, and parks at the census block group (CBG) level using R5R. We calculated activity participation as number of daily activities per person without double counting activities of the same purpose which occur at the same location, following the method used by Allen & Farber (2020).



Accessibility measure:
R5R functions used:
accessibility(),
travel_time_matrix(),
cost_to_closest()



Activity participation:
4 person trips, but 2 activities

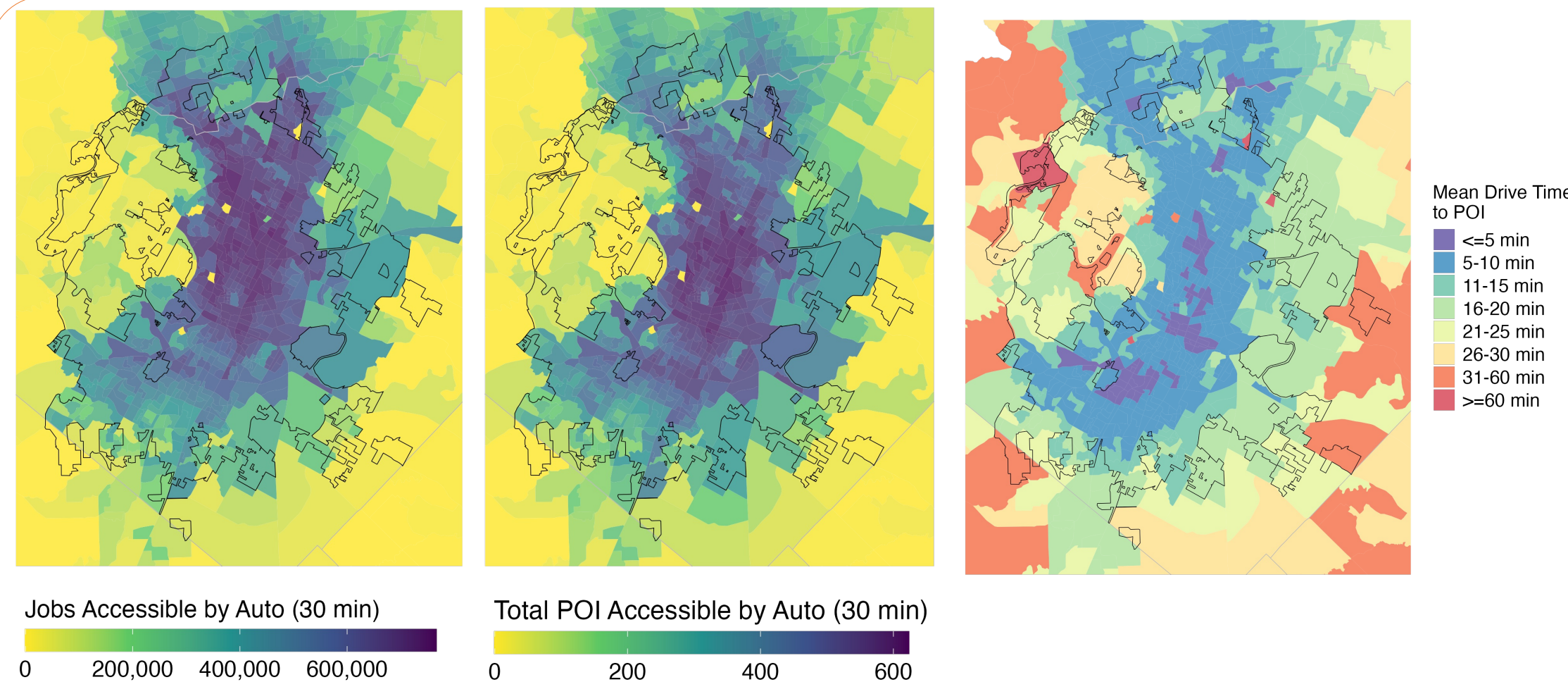
1. Quality of Life. Please rate your satisfaction with the following.	
1. The City of Austin as a place to live	
2. The City of Austin as a place to work	
3. The City of Austin as a place to raise children	
4. The City of Austin as a place to retire	
5. The City of Austin as a place where I feel welcome	
6. Overall quality of life in the city	

Quality of Life measure:
Respondents' satisfaction rating on a scale of 1-5 have been used to calculate average QoL measure

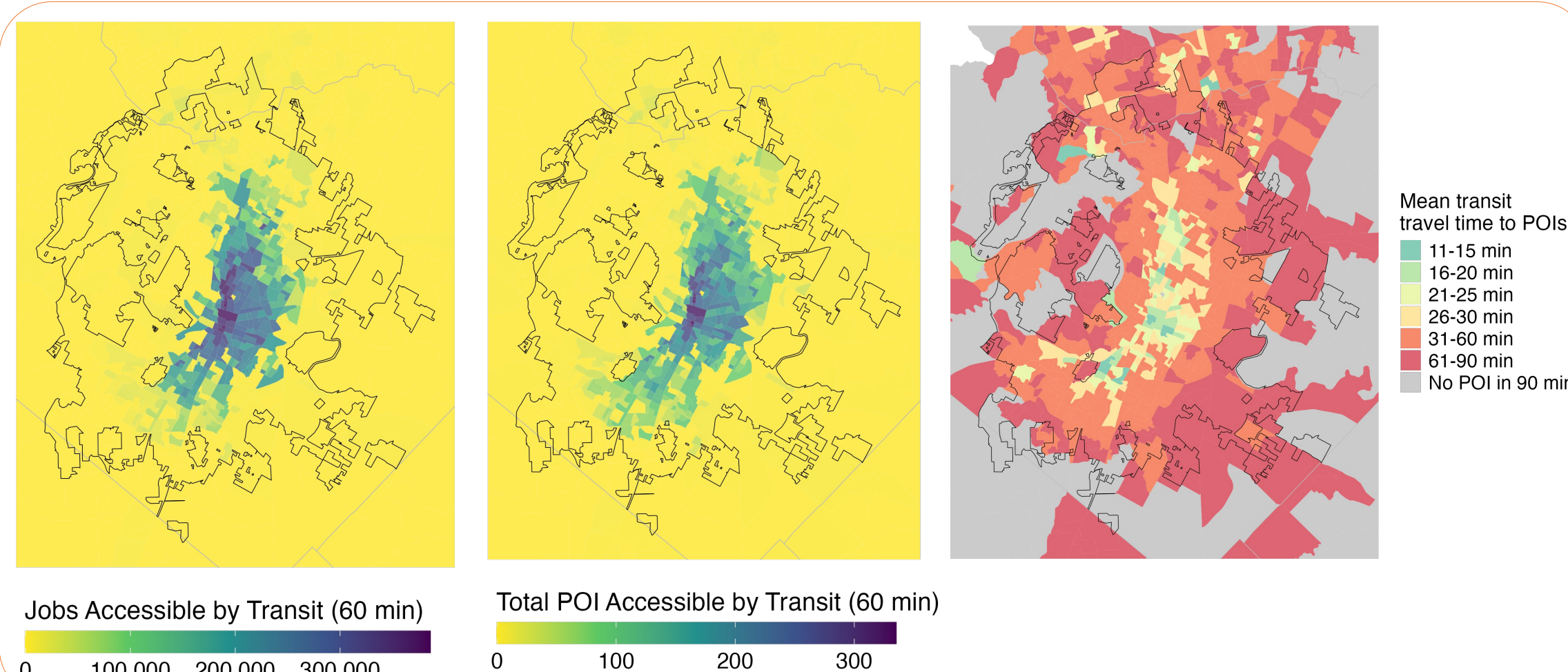
- ❑ **Data Sources:** Jobs data from 2020 LEHD, HIFLD Geo-platform for POI, CBG-level demographic data from ACS 2016-2020, 2017 Capital Area MPO Household Travel Survey data for activity trip calculation, the 2018 City of Austin Community Survey data to evaluate quality of life.
- ❑ **Primal Accessibility:** counting how many opportunities can be reached in a given travel time (e.g., 30 min of car or 60 min of transit). **Dual Accessibility:** measuring the time required to reach N closest destinations of a certain type.
- ❑ The Palma Ratio measures the average accessibility of the wealthiest 10% divided by the average accessibility of the poorest 40%. The Gini Index is a measure of inequality in the distribution of accessibility. The Concentration Index measures the extent to which accessibility inequalities are systematically associated with individuals' socioeconomic levels and represents a superior measure to the Gini coefficient in the context of transport inequalities.
- ❑ Composite Accessibility Index formulation: Expanding USDOT's Equitable Community Explorer method, we calculated transport access sub-component variables scores at granular scale (at block group scale). Variables were normalized, ranked by percentiles, and weighted differently for final index formulation.

Results

Auto Primal and Dual Accessibility



Transit Primal and Dual Accessibility

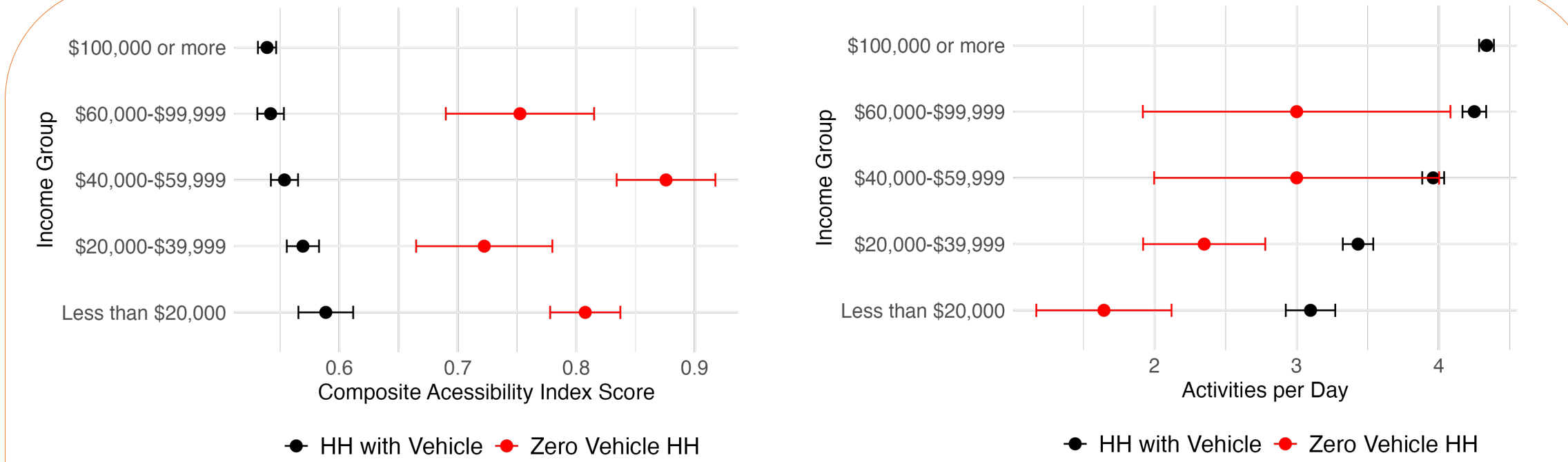


Composite Accessibility Index

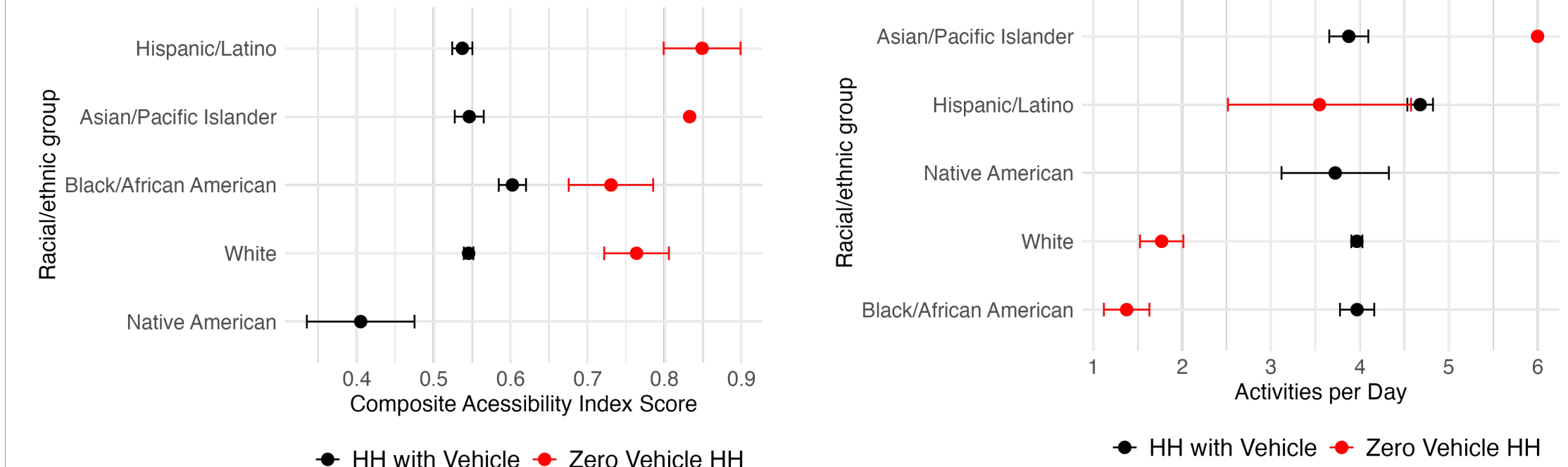


The darkest regions on the map, primarily located in central Austin, indicate areas with the highest composite accessibility index. These areas benefit from a combination of factors such as proximity to jobs, ease of access to points of interest (POIs), and lower average travel times both by car and transit.

Accessibility and Activity Participation disparities by Race/Ethnicity and Income



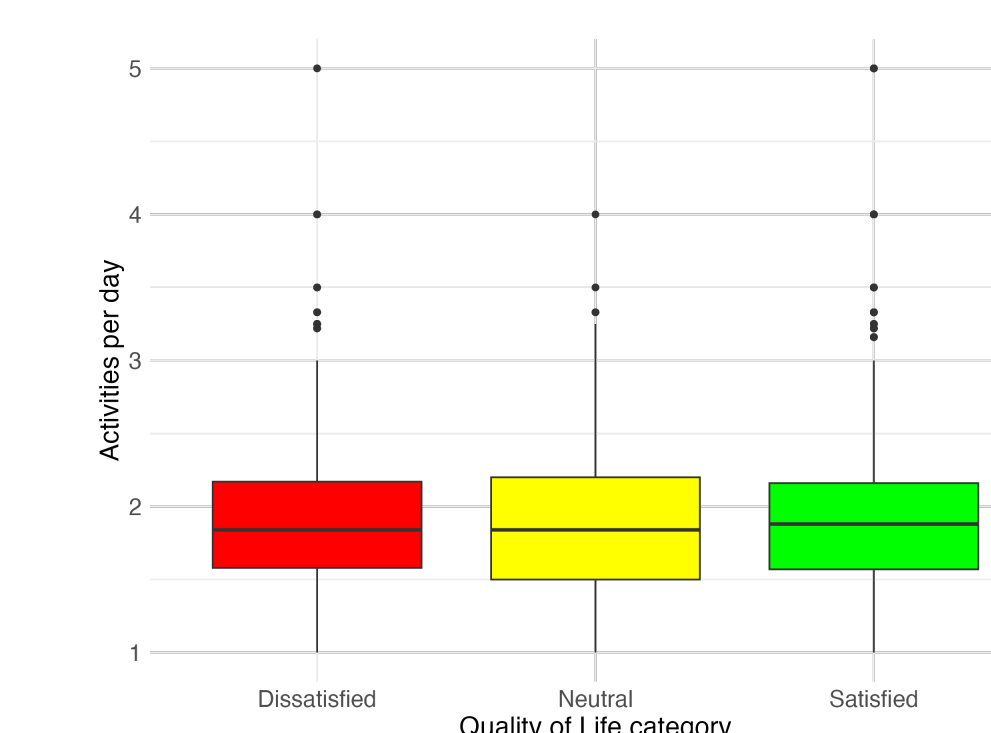
- ❖ Despite higher accessibility scores, lower-income households without vehicles participate in fewer activities per day compared to higher-income households. This suggests that while they have access, other barriers might limit their activity participation.



- ❖ Black/African American households, despite moderate accessibility scores, engage in fewer activities per day, especially those without vehicles. Asian/Pacific Islander and Native American households, with the lowest accessibility scores, also have the lowest activity levels.

Relationship: Access, Activities and QoL

Dependent: Activities per day				
Predictors	Estimates	std. Error	Statistic	P (sig)
(Intercept)	0.36	0.22	1.64	0.101
Income	0.01	0.00	2.00	0.046
Age (0-4)	-0.51	0.10	-5.08	<0.001
Age (5-18)	1.58	0.05	32.17	<0.001
Adult	-0.51	0.08	-6.50	<0.001
Vehicle per Adult	0.25	0.10	2.41	0.016
Black	-0.22	0.16	-1.35	0.177
Composite Accessibility Index Score	0.36	0.16	2.25	0.025
Observations	2796			
R ² / R ² Adjusted	0.378/0.376			



There is no significant difference in the median activity levels across the QoL categories. Individuals' daily activity participation does not vary much based on their reported quality of life.

Key Findings

Accessibility Inequality

Palma Ratio	Gini Index	Concentration Index
0.671	0.29	-0.045

- ❑ The Palma Ratio (0.671) and Concentration Index (-0.045) both indicate that the accessibility distribution slightly favors the poorer population. A Gini Index value of 0.29 indicates a relatively low level of inequality in accessibility distribution.
- ❑ In line with previous findings, we found that in Austin, disadvantaged groups such as low-income and carless households enjoy better accessibility compared to wealthier segments. This advantage is due to their proximity to superior transit services and major employment centers, often found in central locations. Despite lower car ownership, these groups benefit from living in areas with higher accessibility.
- ❑ Households without vehicles across all racial/ethnic and income groups tend to engage in fewer activities per day compared to those with vehicles.
- ❑ Model results indicate that the Composite Accessibility Index Score has a positive and significant impact on activities per day. Specifically, for each unit increase in the Composite Accessibility Index Score, the number of activities per day increases by 0.36, holding all other variables constant.

Conclusion

- ❑ The clear accessibility gap between car and transit modes in Austin underscores the need for improved public transportation infrastructure, especially around sprawling suburban areas.
- ❑ Residents in central Austin, while experiencing better accessibility to urban facilities, find that their ability to engage in daily activities largely depends on their mode of transportation, particularly for those who rely on transit.
- ❑ The study aimed to find a sufficient accessibility threshold where people's activity trips are adequate, reflecting positively on their quality of life score. However, this threshold was not conclusively found, authors are working for more detailed exploration.
- ❑ Further exploration is needed to understand the nuances of transportation-specific life satisfaction and how it impacts overall life satisfaction and well-being.

References

- ❑ Pereira, R. H. M., Saraiva, M., Herszenhut, D., Braga, C. K. V., & Conway, M. W. (2021). r5r: Rapid Realistic Routing on Multimodal Transport Networks with R5 in R. *Transport Findings*, 21262.
- ❑ Martens, K., Singer, M. E., & Cohen-Zada, A. L. (2022). Equity in accessibility: Moving from disparity to insufficiency analyses. *Journal of the American Planning Association*, 88(4), 479-494.
- ❑ Allen, J., & Farber, S. (2020). Planning transport for social inclusion: An accessibility-activity participation approach. *Transportation Research Part D: Transport and Environment*, 78, 102212.
- ❑ Martens, K. (2016). *Transport justice: Designing fair transportation systems*. Routledge.